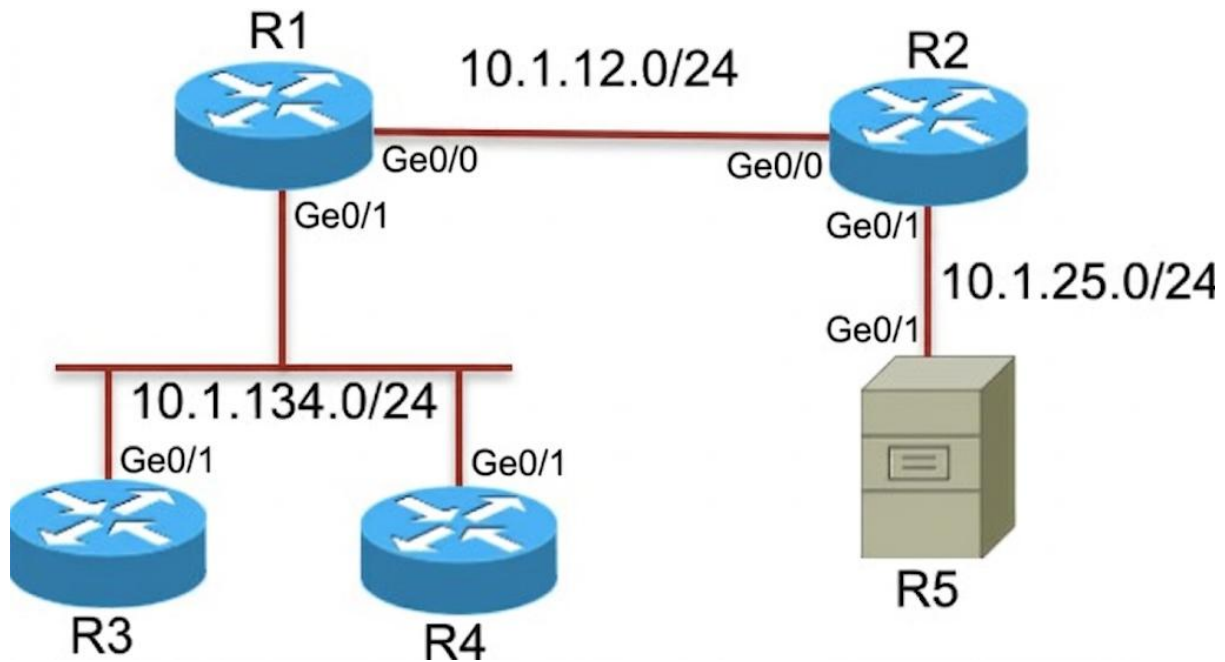


Extended ACL



Tasks

- **Hostnames & IP Addresses:** Configure hostname and IP addresses on all devices using GigabitEthernet interfaces.
- **Routing Protocol:** Configure OSPF 100 on all routers to advertise connected networks in Area 0.
- **Host Emulation:** Disable the routing feature on R5 and configure its default gateway to R2's GigabitEthernet0/1 address (10.1.25.2).
- **Services:** Enable HTTP service on R5. Set the Telnet password to `cisco`.
- **Access Control List:** Configure an Extended ACL on R1 to fulfill the following requirements:
 - Deny HTTP traffic from R3 to R5.
 - Deny Telnet traffic from R4 to R5.
 - Permit all remaining traffic.
- **Verification Target:** At the end of these tasks, R3 and R4 must be able to successfully ping R5.

Overview & Logic

Extended ACLs filter traffic based on Layer 3 (source/destination IP) and Layer 4 (protocol, source/destination ports) information. Unlike standard ACLs, which only inspect source IP addresses, extended ACLs allow granular control over specific application protocols.

In standard network designs, a single server often hosts multiple services. Filtering by server IP address alone would completely isolate the host, blocking all services including basic reachability (ICMP). By using Layer 4 port-based filtering, specific services can be selectively restricted while preserving basic IP connectivity.

In this topology, **R5** acts as an HTTP server with Telnet enabled. **R1** applies the extended access list to enforce traffic policy before traffic routes toward R5.

Configuration

R1 Configuration

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.12.1 255.255.255.0
```

```
no shutdown
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.134.1 255.255.255.0
```

```
ip access-group 100 in
```

```
no shutdown
```

!

router ospf 100

network 10.1.12.0 0.0.0.255 area 0

network 10.1.134.0 0.0.0.255 area 0

!

access-list 100 deny tcp host 10.1.134.3 host 10.1.25.5 eq www

access-list 100 deny tcp host 10.1.134.4 host 10.1.25.5 eq telnet

access-list 100 permit ip any any

!

end

R2 Configuration

enable

configure terminal

hostname R2

!

interface GigabitEthernet0/0

ip address 10.1.12.2 255.255.255.0

no shutdown

!

interface GigabitEthernet0/1

ip address 10.1.25.2 255.255.255.0

no shutdown

!

router ospf 100

network 10.1.12.0 0.0.0.255 area 0

network 10.1.25.0 0.0.0.255 area 0

!

end

R3 Configuration

enable

configure terminal

hostname R3

!

interface GigabitEthernet0/1

ip address 10.1.134.3 255.255.255.0

no shutdown

!

router ospf 100

network 10.1.134.0 0.0.0.255 area 0

!

end

R4 Configuration

enable

configure terminal

```
hostname R4
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.134.4 255.255.255.0
```

```
no shutdown
```

```
!
```

```
router ospf 100
```

```
network 10.1.134.0 0.0.0.255 area 0
```

```
!
```

```
end
```

R5 Configuration

```
enable
```

```
configure terminal
```

```
hostname R5
```

```
no ip routing
```

```
ip http server
```

```
ip default-gateway 10.1.25.2
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.25.5 255.255.255.0
```

```
no shutdown
```

```

!

line vty 0 4

password cisco

login

!

end

```

Verification

Initial State Verification (Before ACL Application)

Verify reachability and service availability from hosts R3 and R4 prior to applying the filtering policies.

R3 Initial Checks

```

t
R3#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP
external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA
external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2 ia - IS-IS inter area, * - candidate
default, U - per-user static route o - ODR, P - periodic downloaded static route, H - NHRP, I -
LISP

```

- replicated route, % - next hop override Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
- O 10.1.12.0/24 [110/65] via 10.1.134.1, 00:28:39, GigabitEthernet0/1
- O 10.1.25.0/24 [110/66] via 10.1.134.1, 00:28:39, GigabitEthernet0/1

```

R3#telnet 10.1.25.5
Trying 10.1.25.5 ... Open
User Access Verification Password:
R5>quit
[Connection to 10.1.25.5 closed by foreign host]

```

```
R3#telnet 10.1.25.5 80
Trying 10.1.25.5, 80 ... Open
```

R4 Initial Checks

``text

```
R4#show ip route ospf
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2 E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2 ia - IS-IS inter area, * - candidate default, U - per-user static route o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP

+ - replicated route, % - next hop override Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

```
O    10.1.12.0/24 [110/65] via 10.1.134.1, 00:26:28, GigabitEthernet0/1
```

```
O    10.1.25.0/24 [110/66] via 10.1.134.1, 00:26:28, GigabitEthernet0/1
```

```
R4#telnet 10.1.25.5
```

```
Trying 10.1.25.5 ... Open User Access Verification
```

Password:

```
R5>quit
```

```
[Connection to 10.1.25.5 closed by foreign host]
```

```
R4#telnet 10.1.25.5 80
```

```
Trying 10.1.25.5, 80 ... Open
```

Both routers successfully establish Telnet and HTTP connections to R5 prior to ACL placement.

Post-ACL Verification

Extended ACLs are applied as close to the source as possible because they can inspect both source and destination details, saving network resources by dropping unauthorized traffic near the ingress point.

Testing Restrictions on R3

```
R3#telnet 10.1.25.5 80
```

Trying 10.1.25.5, 80 ...

% Destination unreachable; gateway or host down

```
R3#telnet 10.1.25.5
```

Trying 10.1.25.5 ... Open

User Access Verification Password:

```
R5>quit
```

[Connection to 10.1.25.5 closed by foreign host]

R1 ACL Counter Check (R3 Test)

```
R1#show access-lists
```

Extended IP access list 100

10 deny tcp host 10.1.134.3 host 10.1.25.5 eq www (1 match)

20 deny tcp host 10.1.134.4 host 10.1.25.5 eq telnet

30 permit ip any any (55 matches)

HTTP traffic from R3 hits rule 10 and is dropped, while Telnet traffic is permitted.

Testing Restrictions on R4

```
R4#telnet 10.1.25.5
```

Trying 10.1.25.5 ...

% Destination unreachable; gateway or host down

R4#telnet 10.1.25.5 80

Trying 10.1.25.5, 80 ... Open R4#

R1 ACL Counter Check (R4 Test)

R1#show access-lists

Extended IP access list 100

10 deny tcp host 10.1.134.3 host 10.1.25.5 eq www (1 match)

20 deny tcp host 10.1.134.4 host 10.1.25.5 eq telnet (1 match)

30 permit ip any any (84 matches)

Telnet traffic from R4 hits rule 20 and is dropped, while HTTP traffic is permitted. All unblocked traffic passes through via rule 30.

ICMP Reachability Test

Confirm ICMP reachability to verify that non-restricted protocols remain functional.R3#ping 10.1.25.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.25.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 84/88/96 ms

R4#ping 10.1.25.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.25.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/62/96 ms

Both R3 and R4 maintain full ICMP connectivity to R5 as intended.